

Systems Analysis and Design

Documentation, Deployment and Release management

INTRODUCTION

- The **implementation phase** consists of developing and testing the system's software, documentation, and new operating procedures.
- **Managing the programming process** is the major task of the systems analyst in this phase.
- While programmers work on programming, the systems analyst design a variety of **tests** to ensure that the system does what it was designed to do.
- During this phase, the systems analysts finalize the **system documentation** and develop the **user documentation**.

DEVELOPING DOCUMENTATION



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- There are two fundamentally different types of documentation:
 - ***system documentation*** is intended to help programmers and systems analysts understand the system and enable them to build it or maintain it;
 - ***user documentation*** is designed to help the user operate the system.

On-line documentation

- There are four key strengths of online documentation:
 - Simpler searching
 - Different representation
 - User Interaction
 - less expensive

Documentation Requirements

- General requirements of all software documentation
 - Should provide for communication among team members
 - Should act as an information repository to be used by maintenance engineers
 - Should provide enough information to management to allow them to perform all program management related activities
 - Should describe to users how to operate and administer the system

Documentation Requirements

- In all software projects some amount of documentation should be created prior to any code being written
 - Design docs, etc.
- Documentation should continue after the code has been completed
 - User's manuals, etc.

Types of Documentation

- There are two fundamentally different types of user documentation:
 - *Process Documents*
 - *Product Documents*

Process Documentation

- Used to record and track the development process
 - Planning documentation
 - Cost, Schedule, Funding tracking
 - Schedules
 - Standards
 - Etc.

Process Documentation

- Has a relatively short lifespan
- Only important to internal development process
 - Except in cases where the customer requires a view into this data
- Some items, such as papers that describe design decisions should be extracted and moved into the *product* documentation category when they become implemented

Product Documentation

- Describes the **delivered** product
- Must evolve with the development of the software product
- Two main categories:
 - System Documentation
 - User Documentation

Product Documentation

- System Documentation
 - Describes how the system works, but not how to operate it
- Examples:
 - Requirements Spec
 - Architectural Design
 - Detailed Design
 - Commented Source Code
 - Including output such as Javadoc
 - Test Plans
 - Including test cases

Product Documentation

- User Documentation has two main types
 - End User
 - System Administrator
- In some cases these are the same people
 - The target audience must be well understood!

Product Documentation

There are five important areas that should be documented for a formal release of a software application

1. Functional Description of the Software
2. Installation Instructions
3. Introductory Manual
4. Reference Manual
5. System Administrator's Guide

Document Quality

- Providing thorough and professional documentation is important for any size product development team
- The problem is that many software professionals lack the writing skills to create professional level documents

Document Structure

- All documents for a given product should have a similar structure
 - A good reason for *product standards*
- The IEEE Standard for User Documentation lists such a structure
 - It is a superset of what most documents need
- The authors “best practices” are:
 1. Put a cover page on all documents
 2. Divide documents into chapters with sections and subsections
 3. Add an index if there is lots of reference information
 4. Add a glossary to define ambiguous terms

Standards

- Standards play an important role in the development, maintenance and usefulness of documentation
- Standards can act as a basis for quality documentation
 - But are not good enough on their own
 - Usually define high level content and organization
- IEEE
 - Has a published standard for user documentation
 - Provides a structure and superset of content areas
 - Many organizations probably won't create documents that completely match the standard

Online Documentation

- Either internal to the application or Web based
- Requires rethinking the presentation style since normal “paper” documentation approaches do not carry over well
- Should act as a supplement to paper documentation
- Biggest benefit of Web docs are that they are always current

Document Preparation

- Covers the entire process of creating and formatting a document for publication
 - Author recommends using specialized (and separate) tools for creating and preparing documents
 - This is only important for user documentation
- It is often important to have a professional writer or document publisher evaluate documents before publication to ensure they look good and will carry over to paper well

Document Storage

- Author Recommends (in 2001)
 - File System to store the actual documents
 - Database to store references to the files with metadata to allow searching and referencing
- Today it is probably better to use a content management systems
 - CVS or Subversion
 - Free and Open Source
 - Easy to setup and maintain

Conclusion

- Good overview of documentation
 - Though most documentation “requirements” are based on contract requirements
 - Hard to cover things like that in a paper like this
- Most of the content seemed to be referring to user documentation
 - Design and other similar docs (often times more graphical than textual) were overlooked

System deployment

System deployment

- It is the process that covers post-development activities like-
 - Configuration
 - Release
 - Installation
 - Updating
 - Adapting
 - Reconfiguration
 - Uninstallation

Traditional software deployment methods

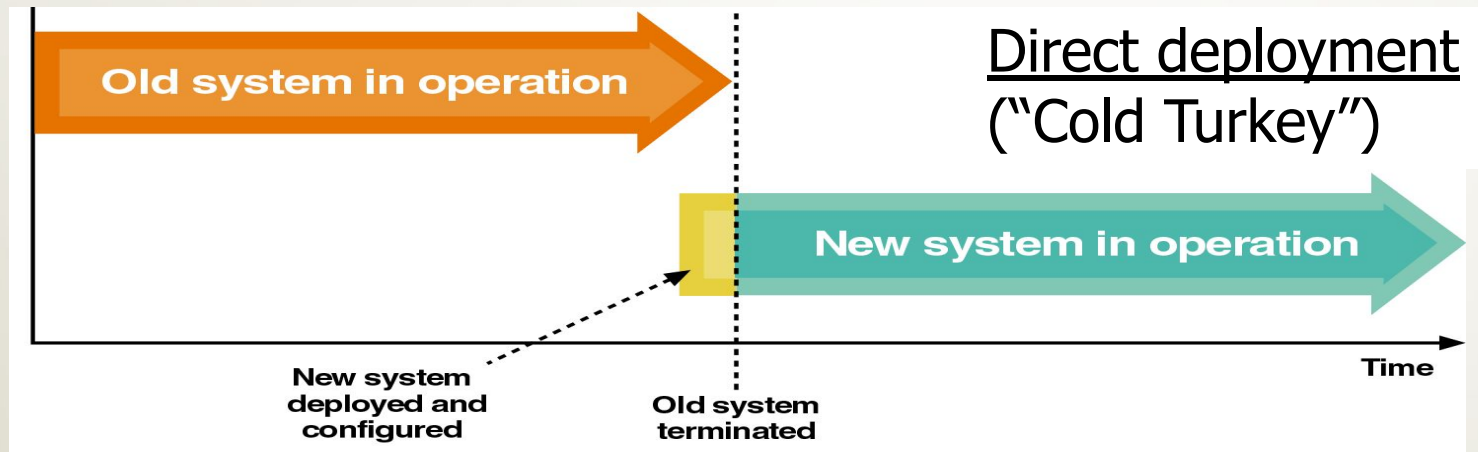


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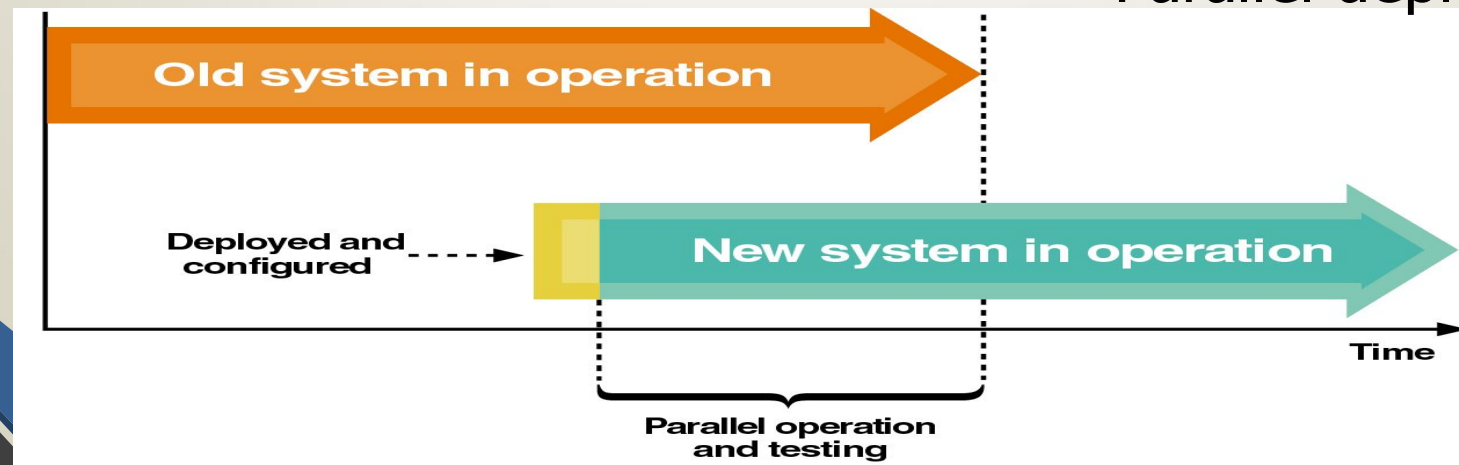
- The foot and hand model
 - Run around on foot and install software by hand.
 - Only viable for small client base.
 - Expensive for the companies.
- The self-service model:
 - The end users install the software themselves.
 - Scales well.
 - Low costs for the company.
 - Becomes difficult as the complexity of installation and configuration increases.

Deployment strategy

Structured methodology

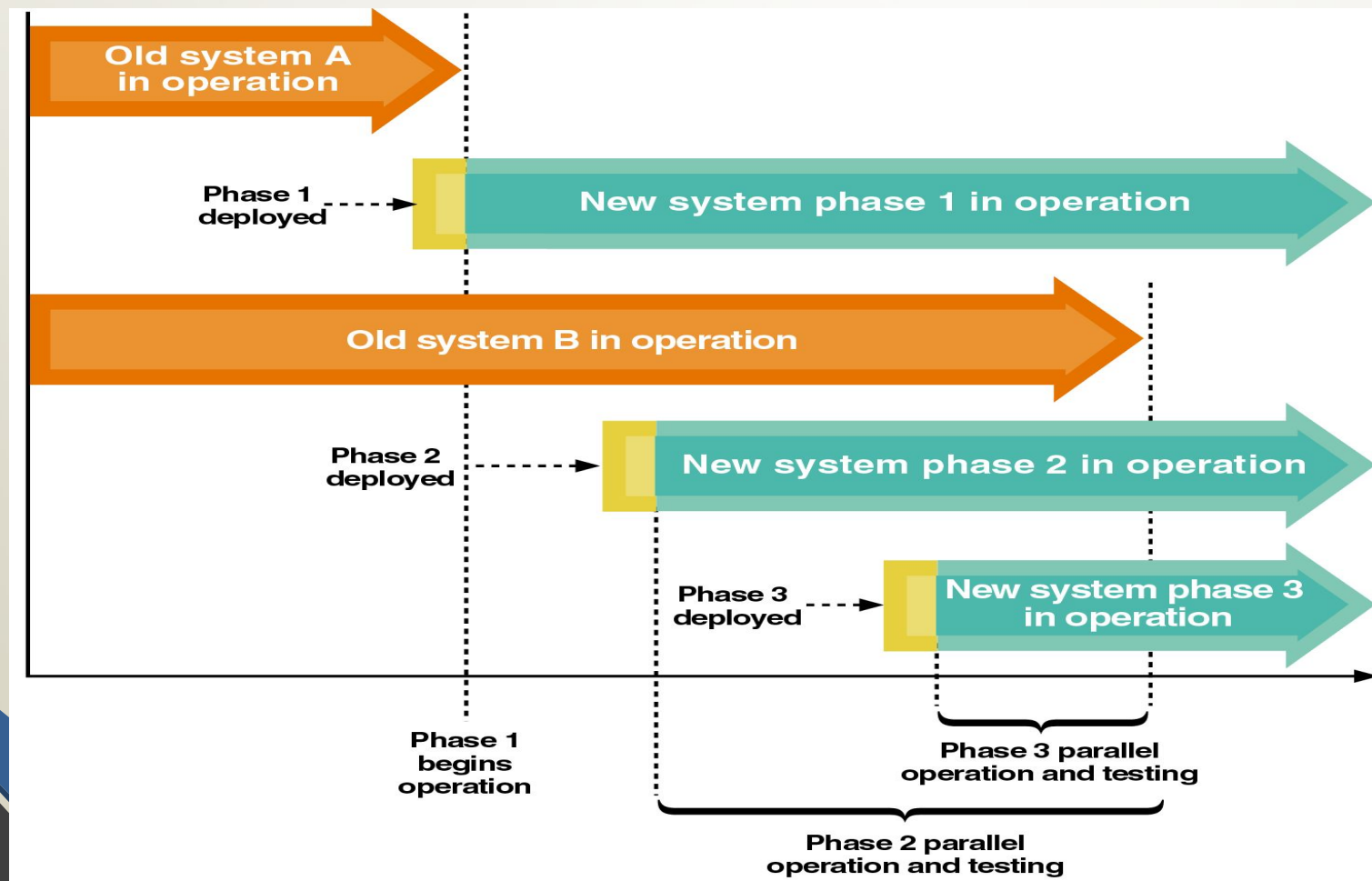


Parallel deployment



Deployment strategy

Phased deployment



Requirements of solution for software deployment



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- Incremental updates
- Versioning
- Automatic installation and configuration
- Centralized Inventory
- Decentralized Control
- Security
- Scalability
- Support for heterogeneous environments
- Live Updates
- Licensing

Software deployment life cycle

- Producer-side processes
 - Release
 - Retire
- Customer-side processes
 - Install
 - Activate
 - Deactivate
 - Reconfigure
 - Update
 - Adapt
 - Remove

Challenges

- Deployment of Windows Software
 - Reliable - installs properly or returns error report
 - Effective - fast, without user intervention and frustration
 - Automated - remotely managed, rebuild easily
 - Targeted - able to add software packages to specific group
 - Upgradeable - able to manage and deploy software upgrades
 - Removable - able to remove software reliably
 - Secure - must expose no new security risks

Possible Solutions

- Manual installation
- RoboCopy
- Active Directory Group Policy Objects
- Third party commercial solutions
- Open Source solutions

Software versioning

- *Alpha* version
 - Test version that is incomplete but ready for some level of testing; dummy data used
- *Beta* version
 - Test version that is stable enough to be tested by end users for an extended period of time; real data used
- *Production* version
 - System version that is formally released in the to users

End of System deployment

Software release management

What is Release Management?

- Release Management is the process of planning, building, testing and deploying hardware and software and the version control and storage of software.
- Its purpose is to ensure that a consistent method of deployment is followed. It reduces the likelihood of incidents as a result of rollouts and ensures that only tested and accepted versions of hardware and software are installed at any time.

Why use Release Management?

- The opportunity to plan expenditure and resource requirements in advance
- A structured approach to rolling out all new software or hardware, which is efficient and effective.
- A changes to software are 'bundled' together for one release, which minimizes the impact of changes on users.
- Testing before rollout
- Ensuring that correct versions are installed at all times

Goals of Release Management

- Release management is geared toward five goals:
 1. Mitigate risk throughout the software delivery life cycle (SDLC)
 2. Align all teams involved in the release
 3. Streamline the development and release processes
 4. Meet deadlines consistently
 5. Align development goals with business goals

The Cycle of Release Management



Release planning

Design

Architecture of a new ICT service. That is the configuration of the pieces of hardware and software involved.

Build

Build is the compilation of components to form the service.

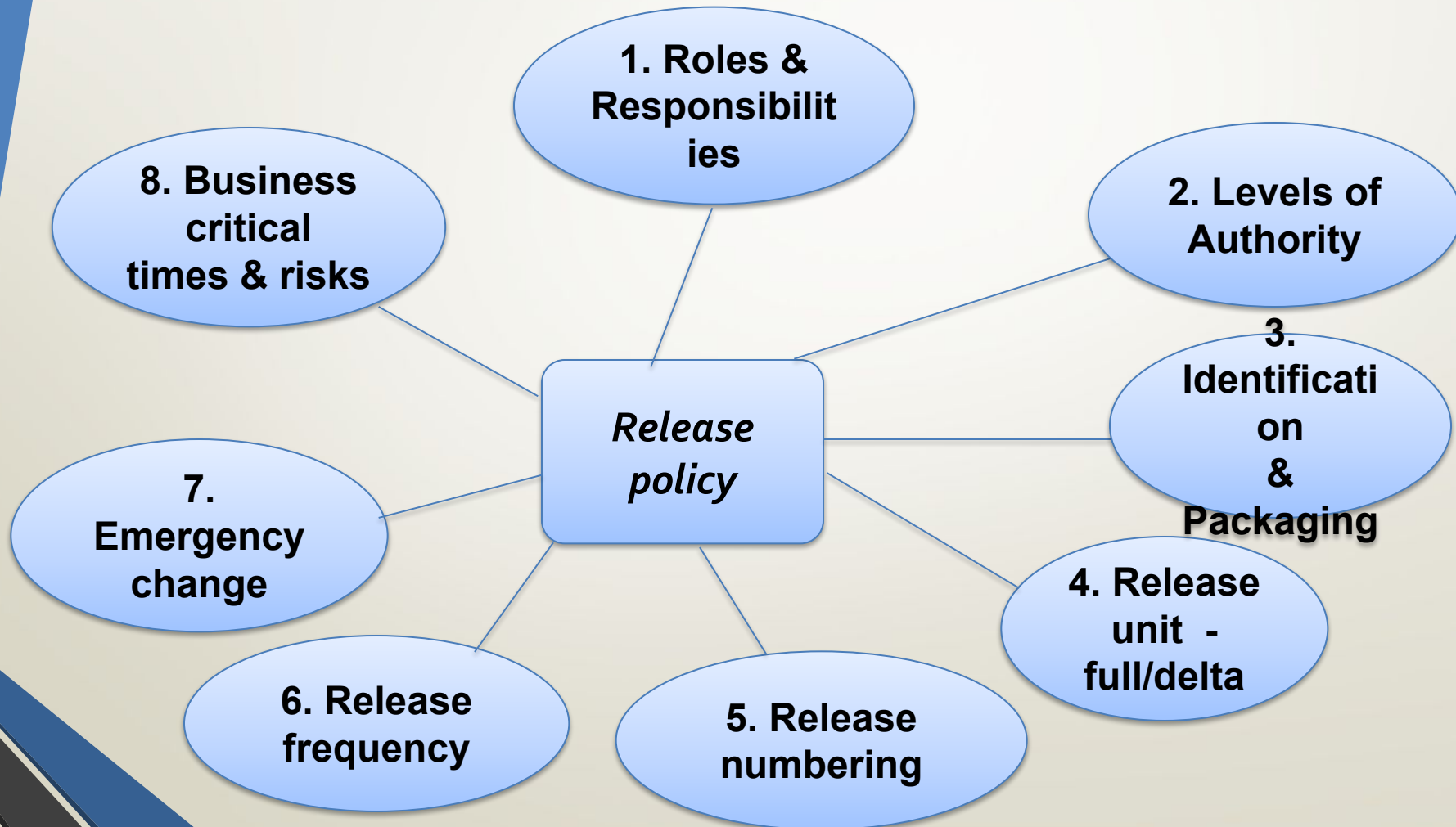
Test

Test is the internal technical support testing.

Acceptance testing

Acceptance testing is functionality testing

Release Policy



Key roles in the release management process



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- **Release manager** - A release manager sets the pace for software rollouts and ensures customers have all the resources they need.
- **Product owner** - Establishing the requirements and acceptance criteria.
- **Quality manager** - Oversees testing to determine whether software passes acceptance criteria without false positives or negatives.
- **DevOps team** - Controls the entire lifecycle of a feature from design to development to production support.

End of Release management